

Experiment No. – 1

Objective: Familiarization of Network Environment, Understanding and using network utilities: ipconfig, netstat, ping, telnet, ftp, traceroute etc.

Theory:

ipconfig: It is a command-line tool used in Windows to display the network configuration of the system. It shows details such as the IP address, subnet mask, and default gateway for each network interface on your machine.

\$ ipconfig

```
PS C:\Users\Kritika Dhiman> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d169:4340:63ac:fa21%23
    IPv4 Address. . . . . : 172.19.50.70
    Subnet Mask . . . . . : 255.255.224.0
    Default Gateway . . . . . : 172.19.32.1

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
PS C:\Users\Kritika Dhiman>
```

netstat: It is a command-line utility that displays network-related information, such as network connections, routing tables, and network interface statistics. It's useful for troubleshooting TCP/IP problems and managing and monitoring your system's interfaces, routes, and connections.

\$ netstat

```
Connection-specific DNS Suffix . : 
PS C:\Users\Kritika Dhiman> netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:49676           LAPTOP-PFDQS03B:49677  ESTABLISHED
TCP    127.0.0.1:49677           LAPTOP-PFDQS03B:49676  ESTABLISHED
TCP    127.0.0.1:49678           LAPTOP-PFDQS03B:49679  ESTABLISHED
TCP    127.0.0.1:49679           LAPTOP-PFDQS03B:49678  ESTABLISHED
TCP    172.19.50.70:49282       whatsapp-cdn-shv-02-bom5:https ESTABLISHED
TCP    172.19.50.70:49417       4.213.25.242:https      ESTABLISHED
TCP    172.19.50.70:49938       52.123.164.130:https    ESTABLISHED
TCP    172.19.50.70:50616       52.123.253.77:https     ESTABLISHED
TCP    172.19.50.70:51108       65:https                ESTABLISHED
TCP    172.19.50.70:51226       nv-in-f94:https         ESTABLISHED
TCP    172.19.50.70:51227       52.123.226.18:https     ESTABLISHED
TCP    172.19.50.70:51228       52.110.14.171:https     ESTABLISHED
TCP    172.19.50.70:51229       40.126.18.33:https      ESTABLISHED
TCP    172.19.50.70:51230       52.111.224.7:https      ESTABLISHED
TCP    172.19.50.70:51231       52.110.14.160:https     ESTABLISHED
TCP    172.19.50.70:51232       52.110.1.195:https      TIME_WAIT
TCP    172.19.50.70:51233       52.110.14.160:https     ESTABLISHED
TCP    172.19.50.70:51234       40.104.77.130:https     ESTABLISHED
TCP    172.19.50.70:51235       40.104.77.130:https     ESTABLISHED
TCP    172.19.50.70:51236       40.104.77.130:https     TIME_WAIT
TCP    172.19.50.70:51237       40.104.77.130:https     TIME_WAIT
TCP    172.19.50.70:51238       40.104.77.130:https     ESTABLISHED
TCP    172.19.50.70:51239       40.104.77.130:https     TIME_WAIT
TCP    172.19.50.70:51240       40.104.77.130:https     TIME_WAIT
TCP    172.19.50.70:51241       20.190.175.24:https     TIME_WAIT
```

pathping: Combination of tracer and ping. It shows the route taken by packets to a destination and measures packet loss and latency at each hop. It is useful for finding network problems, slow connections, and packet loss.

\$pathping <IP address/domain name>

```
PS C:\Users\Kritika Dhiman> pathping google.com

Tracing route to google.com [142.250.193.142]
over a maximum of 30 hops:
 0  LAPTOP-PFDQS03B [172.19.50.70]
 1  172.19.32.1
 2  45.116.207.65
 3  undefined.hostname.localhost [45.248.94.45]
 4  72.14.194.150
 5  tzdelb-at-in-f14.1e100.net [142.250.193.142]

Computing statistics for 125 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
    0          Lost/Sent = Pct  Lost/Sent = Pct  LAPTOP-PFDQS03B [172.19.50.70]
    1    0ms      0/ 100 =  0%     0/ 100 =  0%     |
    2    1ms      0/ 100 =  0%     0/ 100 =  0%     |
    3   12ms      0/ 100 =  0%     0/ 100 =  0%     |
    4   ---      100/ 100 =100%   100/ 100 =100%   72.14.194.150
    5   13ms      0/ 100 =  0%     0/ 100 =  0%     tzdelb-at-in-f14.1e100.net [142.250.193.142]

Trace complete.
```

Ping: It tests connectivity between two networked devices using ICMP (Internet Control Message Protocol). It Measures round-trip time and packet loss between source and destination.

\$ping

```
C:\Windows\System32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name

Options:
  -t          Ping the specified host until stopped.
              To see statistics and continue - type Control-Break;
              To stop - type Control-C.
  -a          Resolve addresses to hostnames.
  -n count    Number of echo requests to send.
  -l size     Send buffer size.
  -f          Set Don't Fragment flag in packet (IPv4-only).
  -i TTL      Time To Live.
  -v TOS      Type Of Service (IPv4-only. This setting has been deprecated
              and has no effect on the type of service field in the IP
              Header).
  -r count    Record route for count hops (IPv4-only).
  -s count    Timestamp for count hops (IPv4-only).
  -j host-list Loose source route along host-list (IPv4-only).
  -k host-list Strict source route along host-list (IPv4-only).
  -w timeout  Timeout in milliseconds to wait for each reply.
  -R          Use routing header to test reverse route also (IPv6-only).
              Per RFC 5095 the use of this routing header has been
              deprecated. Some systems may drop echo requests if
              this header is used.
  -S srcaddr  Source address to use.
  -c compartment Routing compartment identifier.
  -p          Ping a Hyper-V Network Virtualization provider address.
  -4          Force using IPv4.
  -6          Force using IPv6.
```

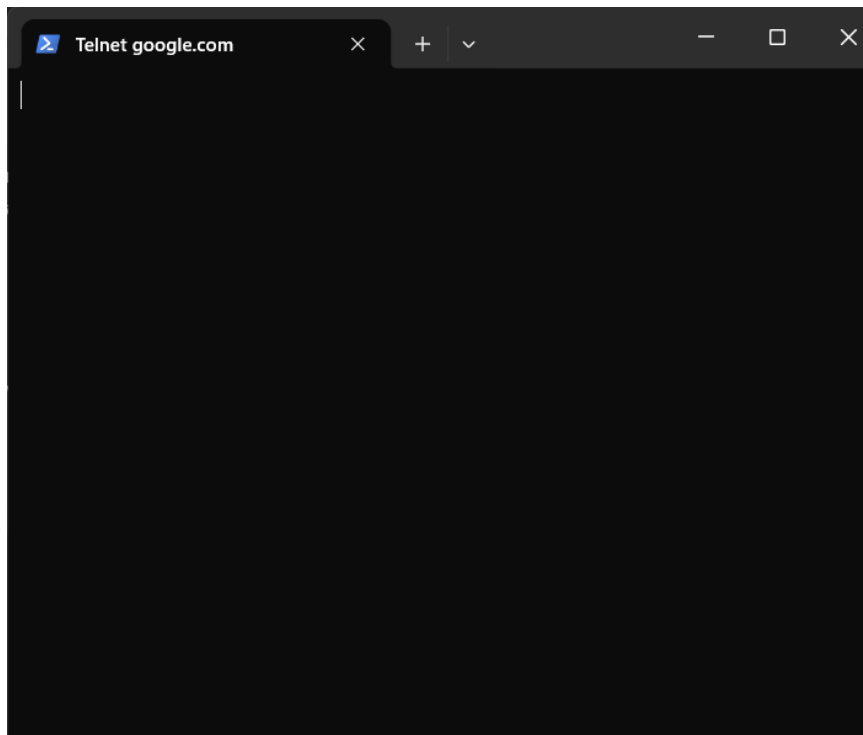
```
C:\Windows\System32>ping google.com

Pinging google.com [2404:6800:4002:830::200e] with 32 bytes of data:
Reply from 2404:6800:4002:830::200e: time=16ms
Reply from 2404:6800:4002:830::200e: time=12ms
Reply from 2404:6800:4002:830::200e: time=15ms
Reply from 2404:6800:4002:830::200e: time=17ms

Ping statistics for 2404:6800:4002:830::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 17ms, Average = 15ms
```

telnet: It is a remote terminal emulator, used to connect to remote systems over TCP/IP networks. Often used to test open ports or log into remote servers.

\$ telnet <IP address/domain name> <port>



ftp: It transfers files between a client and a server over a TCP-based network. Can upload/download files to/from an FTP server.

\$ ftp <IP address/domain name>

```
PS C:\Users\Kritika Dhiman> ftp ftp.dlptest.com
Connected to ftp.dlptest.com.
220 Welcome to the DLP Test FTP Server
200 Always in UTF8 mode.
User (ftp.dlptest.com:(none)): dlpuser
331 Please specify the password.
Password:

230 Login successful.
ftp> dir
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
-rw-r--r--  1 1001    1001      127429 Sep 17 16:30 10.101.1.2_20260917-12310535_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12310675_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12311075_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12311476_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12311957_IVA.jpg
-rw-r--r--  1 1001    1001    128195 Sep 17 16:30 10.101.1.2_20260917-12312778_IVA.jpg
-rw-r--r--  1 1001    1001   134148 Sep 17 16:30 10.101.1.2_20260917-12313619_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12313977_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:30 10.101.1.2_20260917-12314658_IVA.jpg
-rw-r--r--  1 1001    1001   126733 Sep 17 16:31 10.101.1.2_20260917-12315921_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:31 10.101.1.2_20260917-12320139_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:31 10.101.1.2_20260917-12320539_IVA.jpg
-rw-r--r--  1 1001    1001   128823 Sep 17 16:31 10.101.1.2_20260917-12321660_IVA.jpg
-rw-r--r--  1 1001    1001         0 Sep 17 16:31 10.101.1.2_20260917-12322020_IVA.jpg
-rw-r--r--  1 1001    1001   135481 Sep 17 16:31 10.101.1.2_20260917-12323102_IVA.jpg
```

tracert: It is a network diagnostic tool used to track the path data packets take from your computer to a remote host across the internet. It shows each hop along the way (routers or network nodes), along with the response time from each hop. This can help identify where delays or failures occur in a network connection.

\$tracert <IP address/domain name>

```
PS C:\Users\Kritika Dhiman> tracert google.com

Tracing route to google.com [142.250.118.101]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms    172.19.32.1
  2     1 ms     1 ms     1 ms    static.ill.117.250.3.122.bsnl.co.in [117.250.3.122]
  3      *      *        *      Request timed out.
  4      *      *        *      Request timed out.
  5     8 ms     7 ms     7 ms    142.250.164.136
  6     6 ms     6 ms     8 ms    142.251.77.23
  7    12 ms    10 ms     9 ms    192.178.83.226
  8     7 ms     7 ms     7 ms    192.178.83.95
  9     8 ms     7 ms     7 ms    142.251.71.202
 10      *      *        *      Request timed out.
 11      *      *        *      Request timed out.
 12      *
```

getmac: It shows the unique physical address assigned to each network interface, such as Ethernet or Wi-Fi. It is useful for identifying network devices and troubleshooting network connections.

\$getmac

```
PS C:\Users\Kritika Dhiman> getmac

Physical Address      Transport Name
=====
F8-3D-C6-61-19-CF    Media disconnected
82-6D-94-3E-38-A7    Media disconnected
N/A                  Hardware not present
00-E0-92-09-39-FC    \Device\NPF{B044C39D-9BC1-4D60-AF9D-00538CC5AE94}
```

nslookup: It queries Domain Name System(DNS) servers to find the IP address associated with a domain name. It is useful for troubleshooting website connection errors and verifying DNS records.

\$nslookup google.com

```
PS C:\Users\Kritika Dhiman> nslookup google.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4013:807::65
           2404:6800:4013:807::66
           2404:6800:4013:807::64
           2404:6800:4013:807::8a
           142.250.182.238
```

Arp(Address Resolution Protocol): It is used to find the MAC address associated with an IP address on a local network. The arp command displays and modifies the computer's ARP cache/table. It is useful for network troubleshooting and identifying devices on a LAN.

\$arp -a

```
PS C:\Users\Kritika Dhiman> arp -a

Interface: 172.19.50.70 --- 0x17
Internet Address      Physical Address      Type
172.19.32.1           e8-ed-d6-e3-c8-dd    dynamic
172.19.50.54          30-68-93-3d-09-43    dynamic
172.19.50.168         e8-d8-d1-46-8f-8a    dynamic
172.19.51.98          02-d0-f3-fe-05-85    dynamic
172.19.51.187         54-6c-50-1d-11-db    dynamic
172.19.52.68          20-7b-d5-1a-35-5c    dynamic
172.19.63.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Test-NetConnection: It checks if a specific network port is open on a remote server. It is useful for troubleshooting firewall blocks or verifying if a specific service like web or mail is reachable (run this in PowerShell).

\$Test-NetConnection google.com -Port 443

```
PS C:\Users\Kritika Dhiman> Test-NetConnection google.com -Port 443

ComputerName      : google.com
RemoteAddress     : 142.250.193.142
RemotePort        : 443
InterfaceAlias    : Ethernet 8
SourceAddress     : 172.19.50.70
TcpTestSucceeded  : True
```

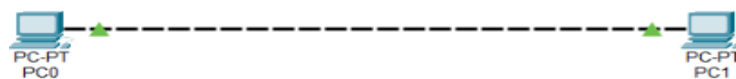
Experiment No. – 2

Objective: Build a simple network topology with routers, switches, and end devices such as PCs or laptops. Configure IP addresses and confirm connectivity between the devices using Packet Tracer.

Procedure: Direct Connection Between End Devices:-Step 1:

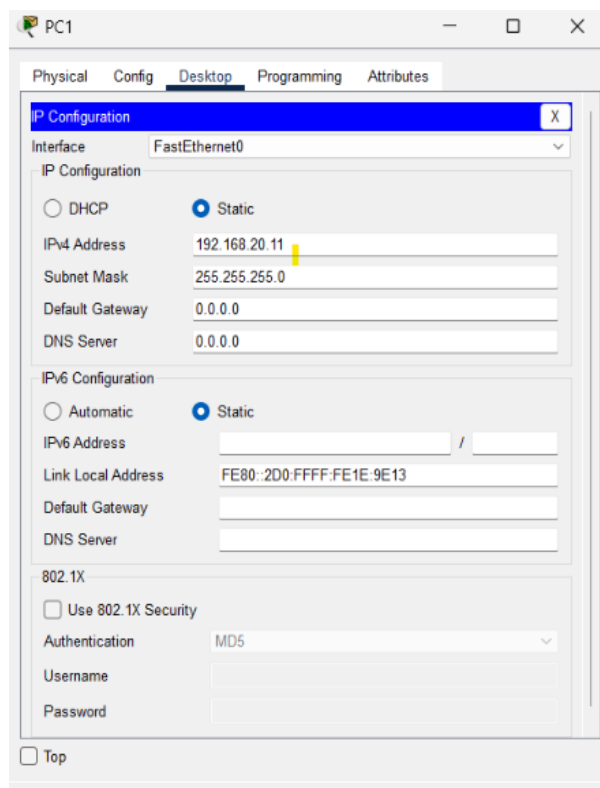
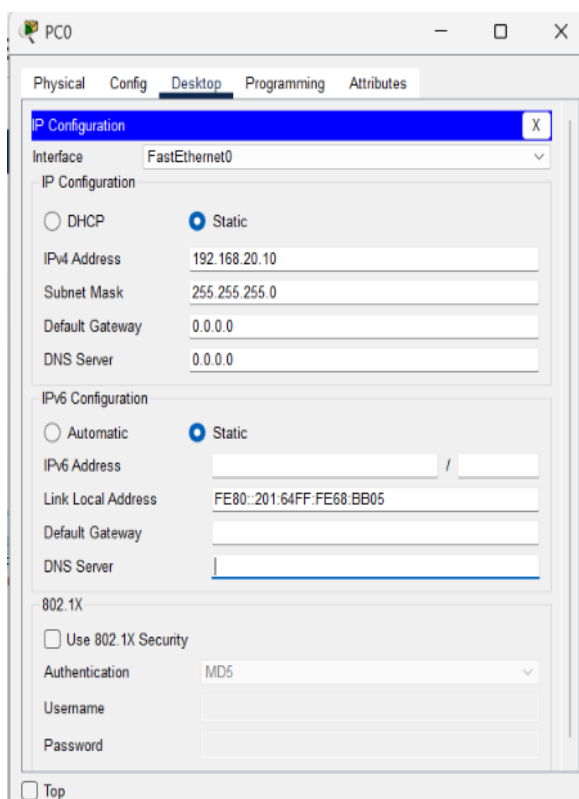
Setting Up Network Devices

- In Cisco Packet Tracer, place the PCs following the topology diagram.
- Connect the devices using crossover cables as needed.



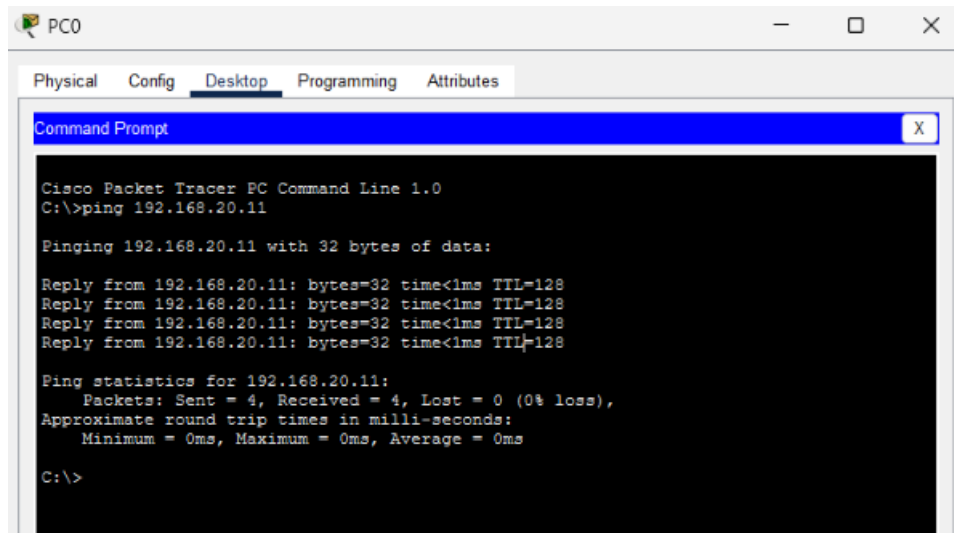
Step2: Configuring IP Addresses on the End Devices

- For each PC, navigate to **Desktop > IP Configuration**.
- Assign the following IP addresses:
 - o PC0: 192.168.20.10
 - o PC1: 192.168.20.11



Step 3: Testing Connectivity

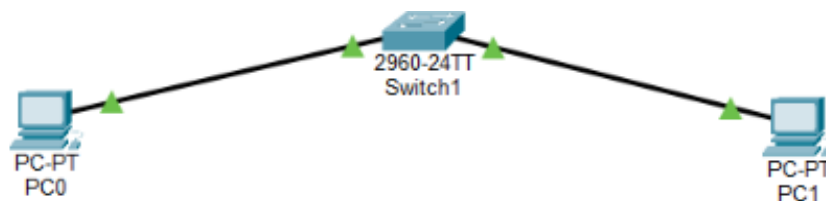
- Open the **Command Prompt** on each PC.
- Use the ping command to test connectivity.
- If the connection is successful, it will resemble the result shown in image.



Direct Connection Between End Devices Using a Switch

Step 1: Setting Up the Devices

- Arrange the PCs and switch in Packet Tracer as shown in image
- Connect the PCs to the switch using straight-through Ethernet cables.

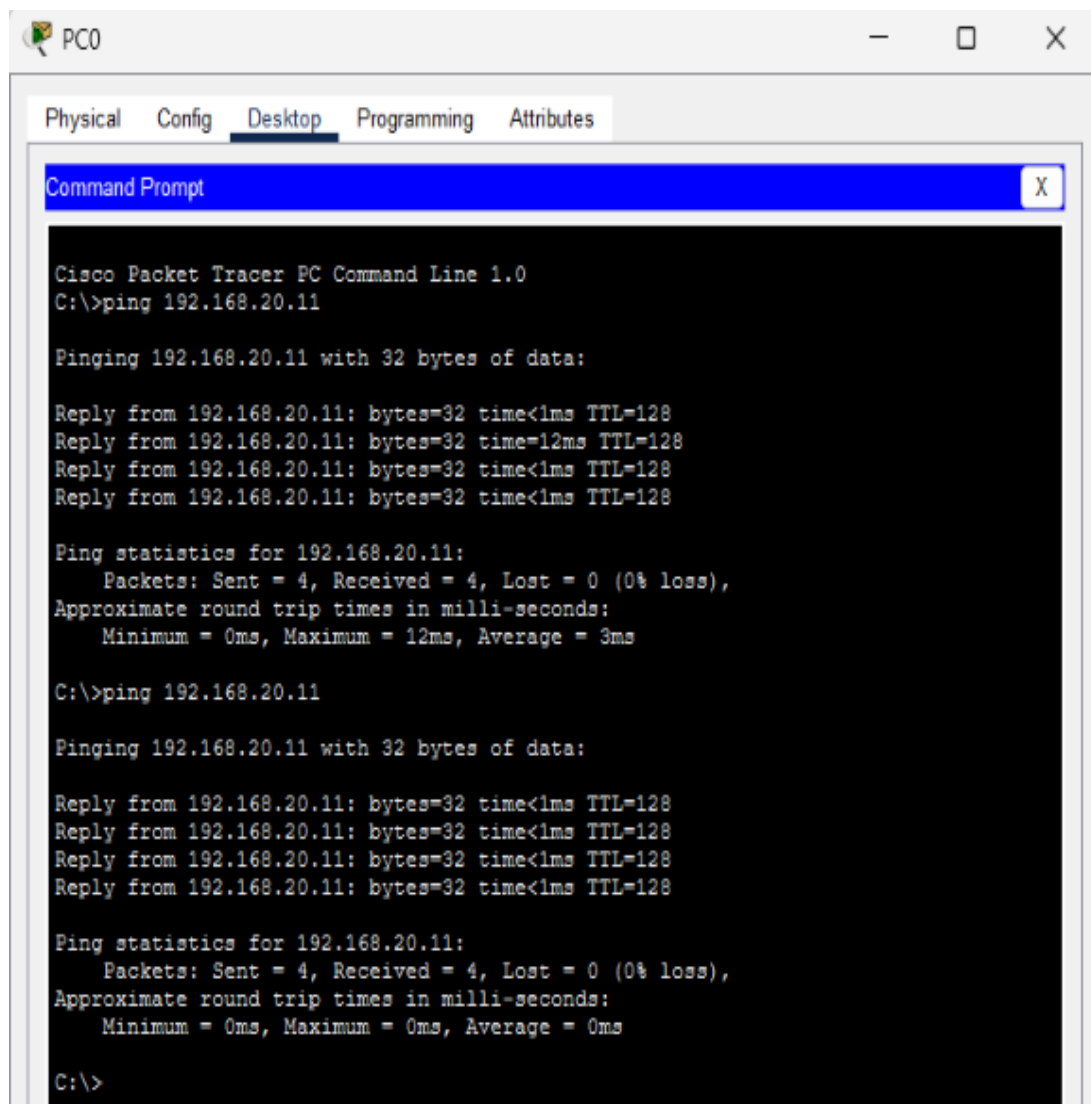


Step 2: Configuring IP Addresses on the PCs

- Assign IP addresses to the PCs as per the previous procedure.

Step 3: Testing Connectivity Between Devices

- On each PC, open the **Command Prompt** and use the ping command to verify successful communication, as shown in image.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of the 'ping 192.168.20.11' command, showing four successful replies with 32 bytes of data, each taking less than 1ms. The ping statistics show 4 packets sent, 4 received, and 0% loss, with round trip times ranging from 0ms to 12ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time=12ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

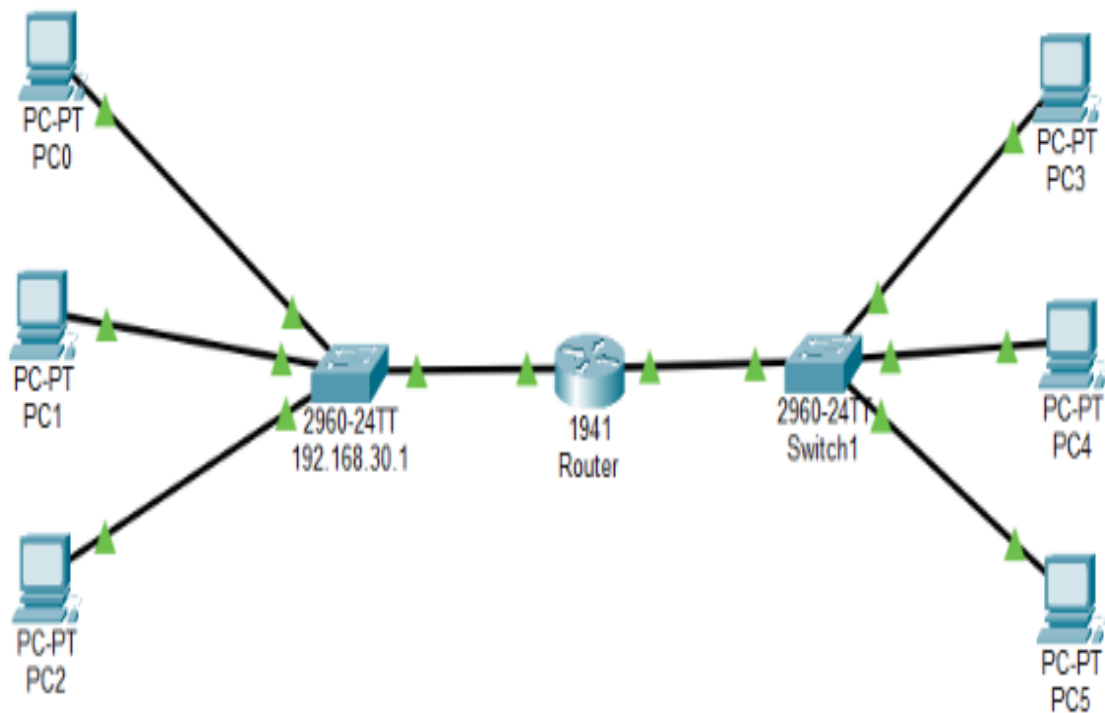
Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Direct Connection Between End Devices Using a Switch and Router:

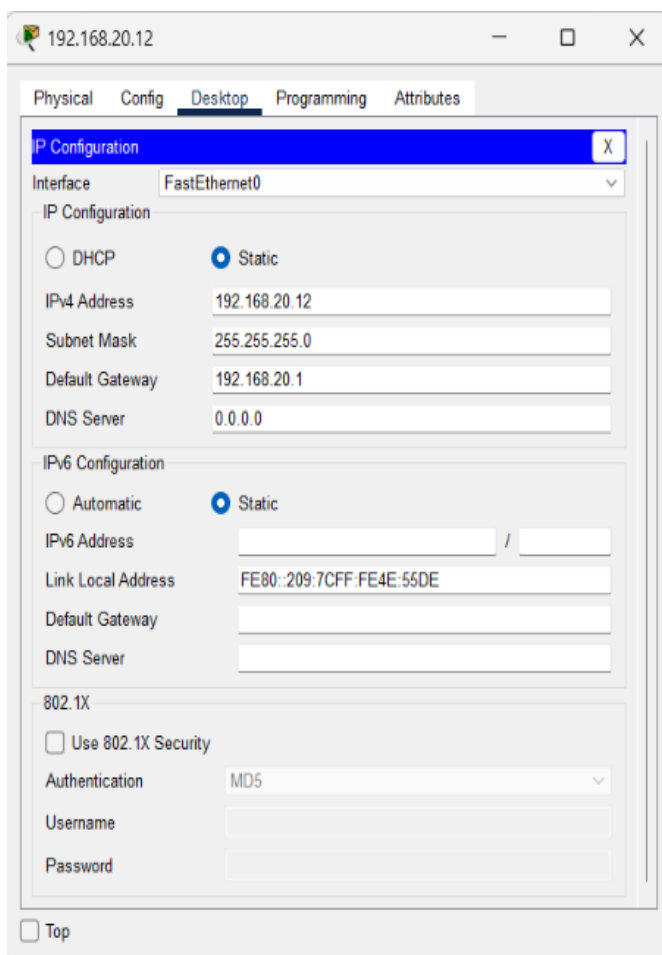
Step 1: Set Up the Devices

- **Arrange the Devices:**
 - In Packet Tracer, position **two LANs** with PCs, switches, and a router, as shown in the image.
 - Use **straight-through Ethernet cables** to connect the PCs to their respective switches, and connect the switches to the router.



Step 2: Configure IPAddresses

- **Assign IPs:**
 - On each PC, go to the **Desktop tab > IP Configuration** and assign IP addresses.
 - Example for PC1: IP = 192.168.30.10, Subnet Mask = 255.255.255.0, Default Gateway = the router's IP for that LAN.
 - Example for PC2: IP = 192.168.30.12, Subnet Mask = 255.255.255.0, Default Gateway = the router's IP for that LAN.



192.168.20.12

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.20.12

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::209:7CFF:FE4E:55DE

Default Gateway:

DNS Server:

802.1X

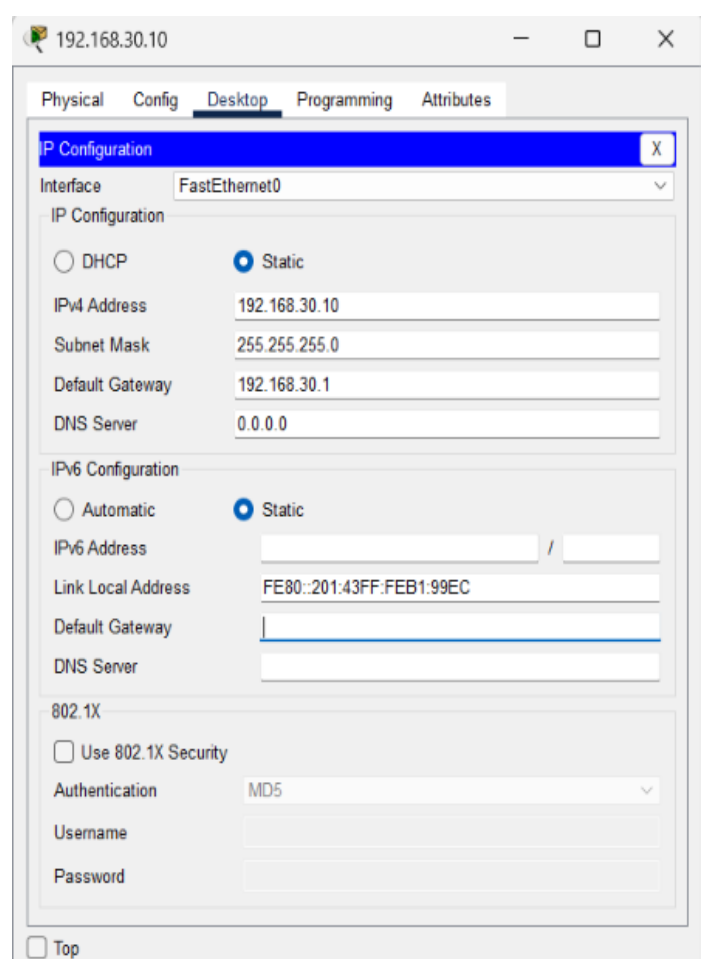
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top



192.168.30.10

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.30.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.30.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:43FF:FEB1:99EC

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Set Default Gateways:

- Ensure the **default gateway** on the PCs matches the corresponding IP of the router's interface.

The screenshot shows the configuration page for the GigabitEthernet0/0 interface. The left sidebar has tabs for Physical, Config, CLI, and Attributes, with 'Config' selected. Under 'Config', there are sub-tabs for GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. The 'INTERFACE' sub-tab is active, showing a list of interfaces with 'GigabitEthernet0/0' selected. The main area displays the configuration for GigabitEthernet0/0: Port Status is 'On', Link Speed is '100 Mbps', Duplex is 'Full Duplex', MAC Address is '0001.C783.E701', IP Configuration shows 'IPv4 Address' as '192.168.20.1' and 'Subnet Mask' as '255.255.255.0', and 'Tx Ring Limit' is '10'. Below the configuration area, a box titled 'Equivalent IOS Commands' contains the following text:

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
```

A 'Top' button is located at the bottom left.

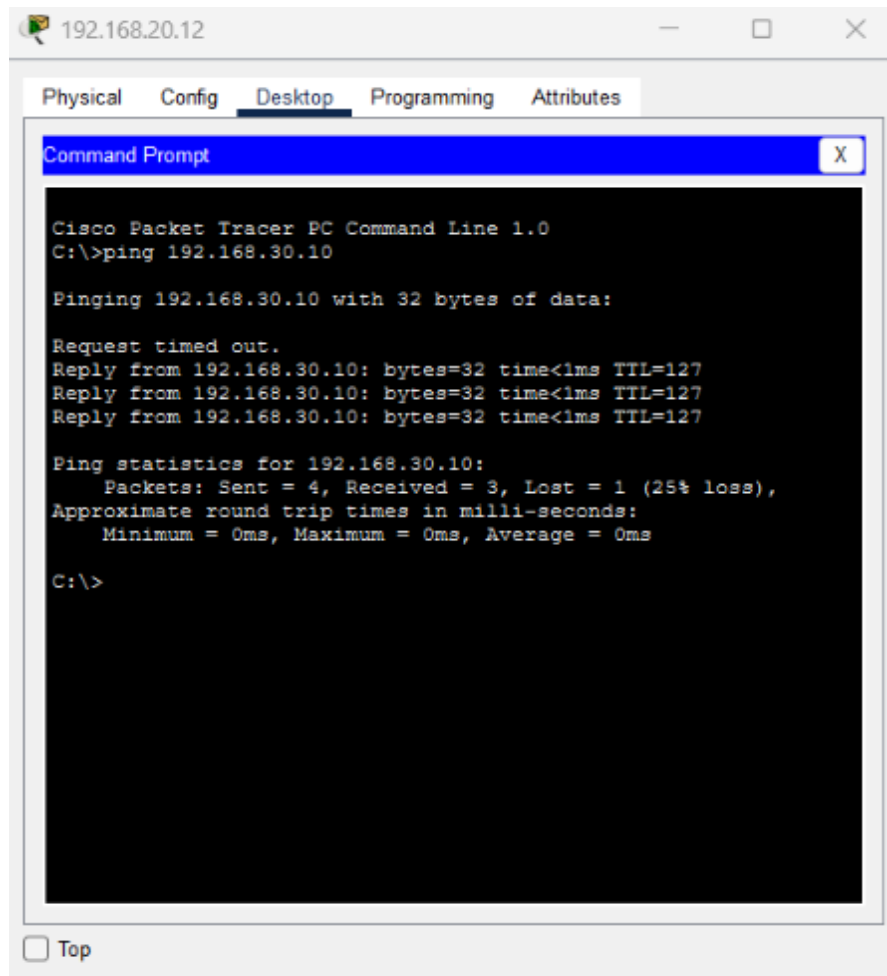
The screenshot shows the configuration page for the GigabitEthernet0/1 interface. The left sidebar is identical to the previous screenshot, with 'Config' selected and the 'INTERFACE' sub-tab active, showing 'GigabitEthernet0/1' selected. The main area displays the configuration for GigabitEthernet0/1: Port Status is 'On', Link Speed is '100 Mbps', Duplex is 'Full Duplex', MAC Address is '00D0.97E2.0702', IP Configuration shows 'IPv4 Address' as '192.168.30.1' and 'Subnet Mask' as '255.255.255.0', and 'Tx Ring Limit' is '10'. Below the configuration area, a box titled 'Equivalent IOS Commands' contains the following text:

```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
ip address 192.168.30.1 255.255.255.0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#
```

A 'Top' button is located at the bottom left.

Step 4: Verify Connectivity:

- **Ping to Test:**
 - Open the **Command Prompt** on **PC1**.
 - Run the ping command to check the connection with a PC on the other LAN.
 - A successful ping will confirm that the two LANs are properly connected.



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC1 (IP: 192.168.20.12). The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of the command 'ping 192.168.30.10'. The output indicates that the ping was successful, with 32 bytes of data being sent and received. The statistics show 4 packets sent, 3 received, and 1 lost (25% loss). The round trip times are all 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

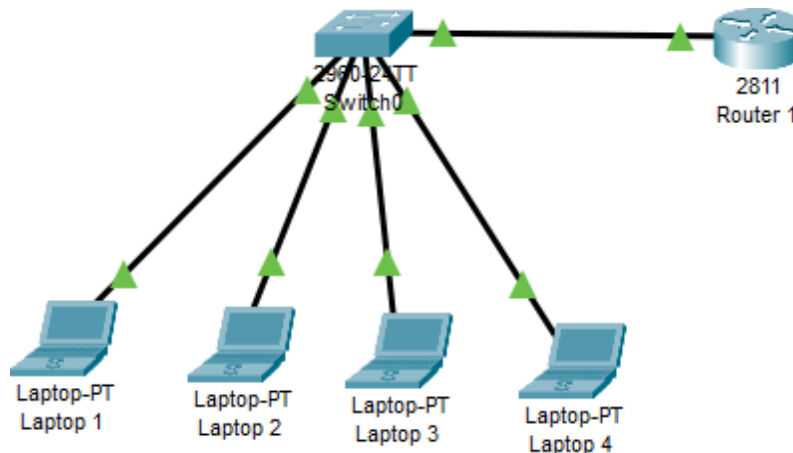

Experiment No. – 3

To configure a DHCP server on a router or a dedicated DHCP server device. Assign IP addresses dynamically to devices on the network and verify successful address assignment. **(Using packet Tracer)**

Procedure:

Case A: Device Setup and Connections (Router as DHCP Server)

1. Open Packet Tracer.
2. **Add Devices:**
 - Place the necessary devices (routers, switches, and end devices) into the workspace according to the layout in the provided diagram (refer to Image).
3. **Connect Devices:**
 - Use straight-through Ethernet cables to link the devices together (PCs to switches, and switches to the router).



Step 2: Configure the Router as a DHCP Server Assign IP address to the router and open CLI in router type following commands.

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.16.10 255.255.255.0
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

```

Router(config-if)#exit
Router(config)#ip dhcp pool firstpool
Router(dhcp-config)#network 192.168.16.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.16.10
Router(dhcp-config)#ip dhcp excluded-address 192.168.16.1 192.168.16.10
Router(dhcp-config)#dns-server 192.168.16.2
Router(dhcp-config)#exit
Router(config)#

```

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.16.10 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp pool firstpool
Router(dhcp-config)#network 192.168.16.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.16.10
Router(dhcp-config)#ip dhcp excluded-address 192.168.16.1 192.168.16.10
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#ip dhcp excluded-address 192.168.16.1 192.168.16.10
Router(config)#dns-server 192.168.16.2
^
% Invalid input detected at '^' marker.

Router(config)#dns-server192.168.16.2
^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 192.168.16.1 192.168.16.10
Router(config)#ip dhcp pool firstpool
Router(dhcp-config)#dns-server 192.168.16.2
Router(dhcp-config)#exit
Router(config)#

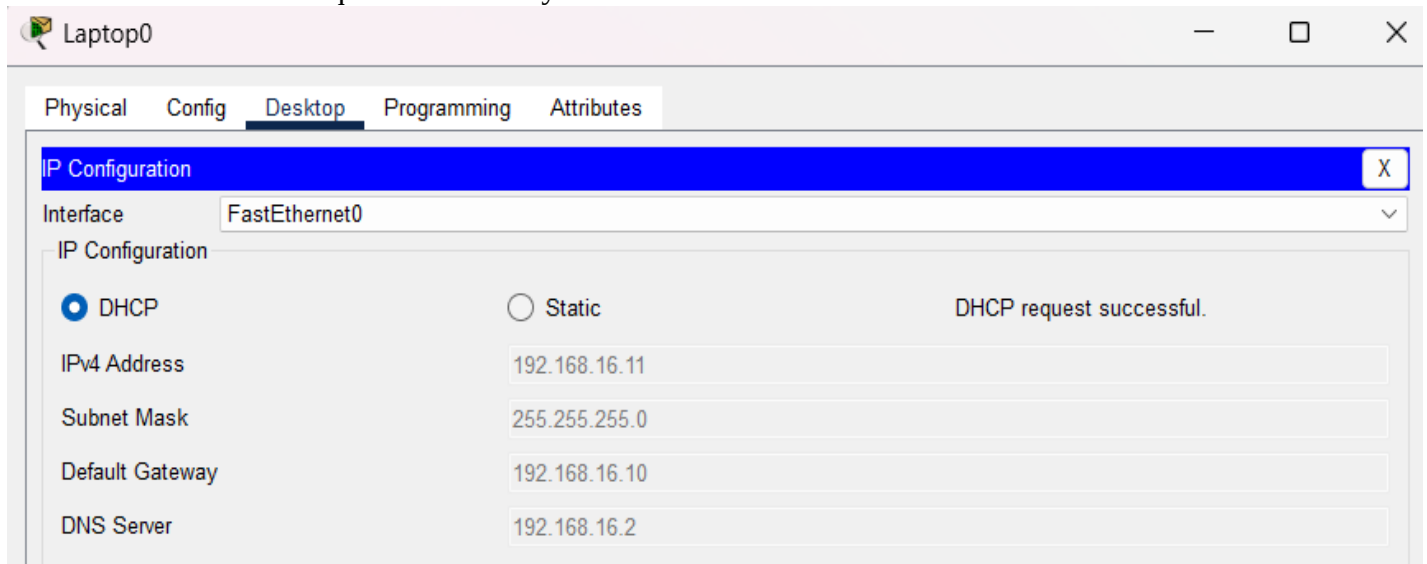
```

Step 3: Verify DHCP Configuration

Now send DHCP request from the end devices (although it is automatically send)

Desktop > ip config > select DHCP

It should return DHCP request successfully.



Step 4: Check for Connectivity

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.16.10

Pinging 192.168.16.10 with 32 bytes of data:

Reply from 192.168.16.10: bytes=32 time<1ms TTL=255
Reply from 192.168.16.10: bytes=32 time<1ms TTL=255
Reply from 192.168.16.10: bytes=32 time<1ms TTL=255
Reply from 192.168.16.10: bytes=32 time<1ms TTL=255

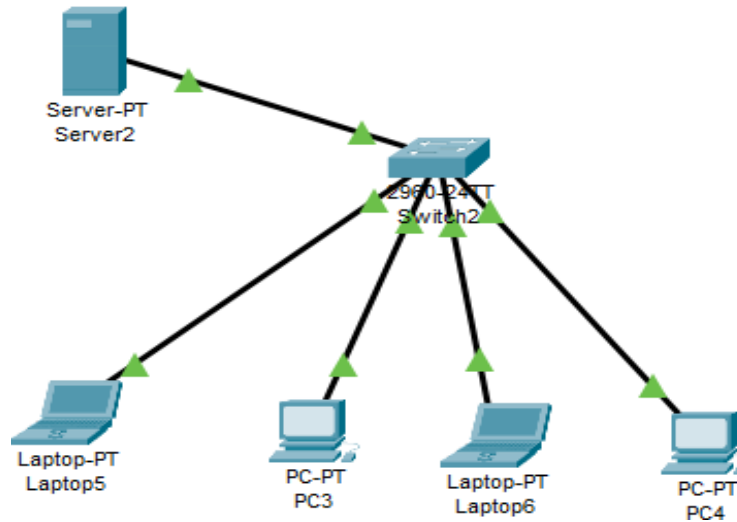
Ping statistics for 192.168.16.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Procedure:

Case B: Device Setup and Connections (As a dedicated DHCP Server)

- Open Packet Tracer.
- **Add Devices:**
 - Place the necessary devices (server, switches, and end devices) into the workspace according to the layout in the provided diagram (refer to Image).
- **Connect Devices:**
 - Use straight-through Ethernet cables to link the devices together (PCs to switches, and switches to the server).



Step 1: Configure the Server as a DHCP Server

Server2

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: lanpool

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.2

Start IP Address: 192 168 1 0

Subnet Mask: 255 255 255 0

Maximum Number of Users: 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
lanpool	192.168.1.1	192.168.1.2	192.168.1.0	255.255....	50	0.0.0.0	0.0.0.0

Step 2: Verify and check for connectivity

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 192.168.1.2

IPv6 Configuration

Step 3: Check for Connectivity

Laptop6

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 4ms

C:\>
```

Experiment No. – 4

Objective: Configure a local DNS server to resolve domain names within a network using Cisco Packet Tracer.

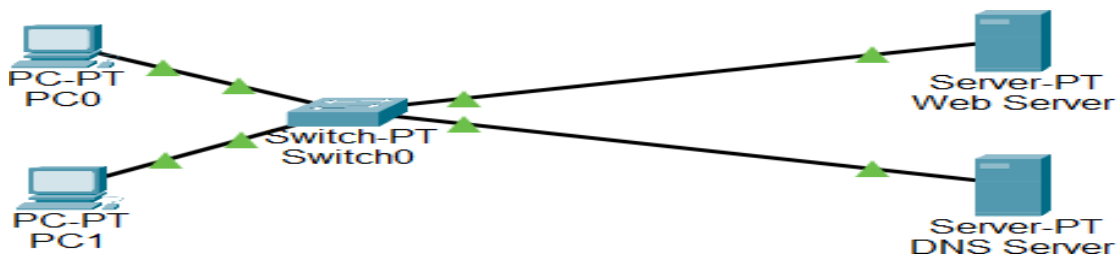
Step1: Connect PC0, PC1, Switch , Web Server and DNS Server using crossover wire.

Step2: Assign static IPs 10.0.0.1 , 10.0.0.2, 10.0.0.3, 10.0.0.4 to PC0 ,PC1 ,Web Server and DNS Server respectively.

Step3: Web Server : Go to the Web server, navigate to Services > HTTP, and modify the index.html file to create a custom website.

Step4: DNS Configuration: On the DNS server, go to Services > DNS, turn the service ON, enter your website URL (e.g., gehu.com), and map it to the Web server's IP address (10.0.0.3). Click Add.

Step5: Client Configuration: On the PC, open IP Configuration and set the DNS Server field to the DNS server's IP address (10.0.0.4)



Web Server

Physical Config Services Desktop Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80:2E0:8FFF:FE48:1E17

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

DNS Server

Physical Config Services Desktop Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.0.0.4

Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80:201:C7FF:FE1C:361E

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

document/d/1kTYkgG5bvM7-wDhiGMQnatsL7bvyum8vwewL5QSLkDDTY/edit?tab=t.0

Web Server

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

File Name: index.html

```
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Document</title>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

File Manager Save

☐ Top

